Notes: Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry (Econometrica, 2018)

Really Uncertain Business Cycles

Contents

1 Big picture

- **Question.** How important are uncertainty shocks for business cycles, and can increases in microeconomic uncertainty generate aggregate recessions in a quantitative general equilibrium model?
- **Empirical starting point.** The paper documents that plant-level and firm-level dispersion rises sharply in recessions. Recessions are not only negative first-moment events; they also feature higher cross-sectional volatility.
- **Core measurement idea.** Establishment-level TFP shocks are constructed from Census microdata. Their cross-sectional dispersion is used as a measure of microeconomic uncertainty.
- **Main empirical facts.** Dispersion of establishment-level TFP shocks, sales growth, output growth, and stock returns is countercyclical. This pattern appears at plant, firm, and industry levels.
- **Model contribution.** The paper builds a DSGE model with heterogeneous firms, time-varying aggregate and idiosyncratic uncertainty, and nonconvex adjustment costs in capital and labor.
- **Main quantitative result.** A pure uncertainty shock generates a sharp temporary fall in output of around 2.5 percent, primarily through real-options inaction, factor misallocation, and delayed adjustment.
- **Need for first moments.** A pure uncertainty shock alone has difficulty matching consumption dynamics. A recession-like simulation requires both a positive second-moment shock and a negative first-moment productivity shock.
- **Policy result.** Uncertainty makes standard first-moment policies, such as wage subsidies, less effective because firms become less responsive when they are far from hiring and investment thresholds.
- **Takeaway.** Business cycles are better described as joint movements in first and second moments: recessions combine bad news about productivity with a rise in uncertainty.

2 Incremental contribution of this paper

2.1 Relative to earlier uncertainty-shock papers

Earlier work, including Bloom (2009), showed that uncertainty shocks can create short-run drops in investment and hiring through real-options effects. This paper extends that literature in several ways:

- it uses large-scale Census microdata to construct direct plant-level measures of uncertainty;
- it connects these micro measures to financial-market proxies such as stock-return volatility;
- it embeds uncertainty in a richer DSGE model with both capital and labor adjustment frictions;
- it estimates the uncertainty process structurally using simulated method of moments;
- it studies policy effectiveness in the presence of uncertainty shocks.

Interpretation: The paper's key advance is to connect micro evidence, macro dynamics, and policy experiments in one framework rather than treating uncertainty as only a reduced-form aggregate shock.

2.2 Relative to standard RBC models

Standard RBC models usually treat business cycles as first-moment movements in aggregate productivity. This paper shows that:

- recessions also involve large increases in cross-sectional dispersion;
- second-moment shocks alone can generate large output, labor, and investment declines;

- first-moment shocks are still needed to match consumption and recession dynamics;
- uncertainty changes the responsiveness of the economy to policy.

Interpretation: The incremental contribution is not to replace technology shocks with uncertainty shocks, but to show that realistic recessions require both.

2.3 Relative to empirical dispersion facts

A growing literature documents countercyclical dispersion in firms, plants, income, and prices. This paper adds:

- a direct Census-based TFP-shock measure at the establishment level;
- evidence that dispersion rises in recessions for sales, output, and stock returns;
- evidence that plant-level shocks correlate with parent-firm stock-return volatility;
- a structural model that translates dispersion into aggregate dynamics.

Interpretation: The paper turns dispersion from a stylized fact into an estimated structural state variable that affects firm decisions and aggregate outcomes.

3 Data and measurement

3.1 Census plant-level data

The empirical analysis uses confidential U.S. Census Bureau manufacturing data. The authors combine the Census of Manufactures and the Annual Survey of Manufactures to follow establishments over time. The plant-level dataset allows the authors to estimate productivity shocks and measure dispersion within years.

- Establishments are observed over repeated years, allowing the paper to remove fixed establishment heterogeneity.
- The analysis focuses on establishments with enough observations to estimate productivity dynamics reliably.
- The core empirical period is 1972–2010 or 1972–2011 depending on the variable and sample.

Interpretation: The Census data are crucial because uncertainty is a microeconomic concept: the relevant object is the dispersion of shocks across establishments, not only aggregate volatility.

3.2 TFP shocks

The authors estimate establishment-level log TFP and then recover shocks from an autoregressive specification:

$$\log z_{j,t} = \rho \log z_{j,t-1} + \mu_j + \lambda_t + e_{j,t}. \tag{1}$$

Here μ_j is an establishment fixed effect, λ_t is a year fixed effect, and $e_{j,t}$ is the residual interpreted as an establishment-level TFP shock.

Interpretation: The residual $e_{j,t}$ captures idiosyncratic plant-level innovations after controlling for permanent plant differences and aggregate year effects. Because revenue-based TFP is used, the residual can contain both productivity and demand shocks.

3.3 Microeconomic uncertainty

The main uncertainty measure is the cross-sectional dispersion of establishment-level TFP shocks within each year:

$$\sigma_t^z = SD_j(e_{j,t}). \tag{2}$$

The paper also considers robust dispersion measures, especially the interquartile range of $e_{j,t}$.

Interpretation: The object is not the level of productivity but the dispersion of productivity innovations. A recession can therefore combine a lower mean and a wider cross-sectional distribution.

3.4 Alternative uncertainty measures

The paper checks that the uncertainty pattern is not specific to one measure.

- It uses sales growth and output growth dispersion.
- It examines dispersion at plant, firm, and industry levels.
- It constructs stock-return volatility measures for publicly traded parent firms.
- It compares Census-based TFP dispersion with measures from Compustat and stock market data.

Interpretation: These robustness exercises show that the countercyclical uncertainty pattern is broad and not an artifact of one data source or one TFP estimation method.

4 Empirical specifications

4.1 Recession regressions

The paper first regresses annual uncertainty measures on recession exposure:

$$Uncertainty_t = \alpha + \beta Recession_t + u_t. \tag{3}$$

The recession variable measures the share of quarters in year t that are NBER recession quarters.

Interpretation: A positive β indicates that uncertainty is higher in recession years. Table I shows that standard deviation and interquartile range of plant-level TFP shocks increase strongly during recessions.

4.2 Industry-year regressions

To show that the pattern holds within industries, the paper estimates:

$$IQR_{i,t} = a_i + b_t + \gamma \Delta y_{i,t} + \varepsilon_{i,t}, \tag{4}$$

where $IQR_{i,t}$ is the dispersion of establishment TFP shocks in industry i and year t, and $\Delta y_{i,t}$ is industry output growth.

Interpretation: The negative coefficient on industry growth means that industries with lower growth have higher within-industry uncertainty.

4.3 Stock-return-based measures

The paper links plant-level uncertainty to parent-firm stock-return volatility. For publicly traded firms, it examines whether the size of plant-level TFP shocks correlates with daily stock-return standard deviation.

Interpretation: This validates high-frequency stock-return volatility as a financial proxy for real microeconomic uncertainty.

5 Model framework

5.1 Heterogeneous firms

The model features a continuum of heterogeneous firms. Firms face idiosyncratic productivity shocks, aggregate productivity shocks, and time-varying uncertainty. They choose labor and capital subject to adjustment frictions.

Interpretation: Heterogeneity matters because uncertainty operates through the distribution of firms relative to adjustment thresholds.

5.2 Adjustment costs

The model includes nonconvex adjustment costs in capital and labor.

- Capital adjustment costs make investment lumpy.
- Labor adjustment costs make hiring and firing costly.
- Firms become inactive when uncertainty rises because waiting has option value.

Interpretation: Without adjustment costs, higher uncertainty can even raise investment through convexity effects. The real-options mechanism requires frictions that make waiting valuable.

5.3 Uncertainty process

The model includes stochastic processes for both aggregate and idiosyncratic uncertainty. High-uncertainty states have higher volatility at the macro and micro levels. These processes are estimated by simulated method of moments.

Interpretation: The model treats uncertainty as a state variable that moves over time and is persistent, rather than as a one-time exogenous shock.

5.4 Simulated method of moments

The authors estimate uncertainty parameters by matching moments of macro and micro volatility:

- mean, standard deviation, skewness, and serial correlation of aggregate volatility;
- mean, standard deviation, skewness, and serial correlation of the micro uncertainty measure.

Interpretation: The SMM procedure anchors the model's uncertainty process in observed micro and macro data.

6 Quantitative mechanisms

6.1 Pure uncertainty shock

A pure uncertainty shock raises the variance of shocks while holding the first moment fixed. The model predicts:

- a sharp drop in output;

- a rapid fall in labor and investment;
- a temporary rebound as uncertainty subsides;
- increased misallocation because productive firms delay expansion and unproductive firms delay contraction.

Interpretation: The fall in output reflects a combination of inaction and misallocation. Firms wait rather than hire, fire, invest, or disinvest.

6.2 Need for a negative first-moment shock

A pure uncertainty shock alone has difficulty matching the behavior of consumption. Consumption can overshoot because households smooth intertemporally and investment falls. To create a more realistic recession, the paper adds a negative first-moment productivity shock.

Interpretation: The recession mechanism is joint: uncertainty creates inaction and misallocation, while a negative first-moment shock generates the fall in consumption and magnifies the output contraction.

6.3 Decomposition of general equilibrium and adjustment costs

The paper compares the baseline general equilibrium model with partial equilibrium versions.

- Without adjustment costs, uncertainty can raise output through the Oi-Hartman-Abel effect.
- With adjustment costs, firms pause adjustment and output falls.
- General equilibrium consumption smoothing affects the rebound and persistence of the cycle.

Interpretation: Adjustment frictions are essential for uncertainty shocks to generate recessions rather than expansions.

6.4 Policy effectiveness

The paper studies a wage subsidy during normal times and during an uncertainty shock. The same first-moment policy is less effective when uncertainty is high.

Interpretation: Uncertainty pushes firms away from action thresholds, so a given policy-induced price change moves fewer firms into hiring or investment.

7 Identification and interpretation

7.1 Empirical identification

The empirical strategy is descriptive but disciplined:

- micro uncertainty is directly measured from plant-level TFP shocks;
- recession timing is external to individual establishments;
- industry-year regressions compare dispersion across industries over time;
- stock-return measures validate the link between real and financial uncertainty proxies.

Interpretation: The paper does not claim that measured uncertainty shocks are the only source of recessions. It shows that uncertainty rises systematically in recessions and that a calibrated model can translate this into aggregate dynamics.

7.2 Structural identification

The model is disciplined by moments of uncertainty and calibrated adjustment costs. The interpretation of simulations depends on the assumption that the model's adjustment frictions and uncertainty processes capture relevant firm-level behavior.

Interpretation: The model is not identified by one regression coefficient. It is identified by the joint ability to match uncertainty moments and business-cycle responses.

8 Tables and figures

This section uses directly cropped visuals from the paper. Nearby figures and tables are placed on the same row where readability allows. Single visuals use width 0.6.

8.1 Figure 1 and Figure 2: distributions widen in the Great Recession

Read:

- Figure 1 shows that the distribution of establishment-level TFP shocks becomes wider during the Great Recession.
- Figure 2 shows an even stronger widening for sales growth rates.
- The figures visually establish the main empirical fact: recessions are high-dispersion periods.

Economic message: The Great Recession is not only a leftward shift in mean outcomes. It also features a large increase in the cross-sectional variance of firm and plant shocks.

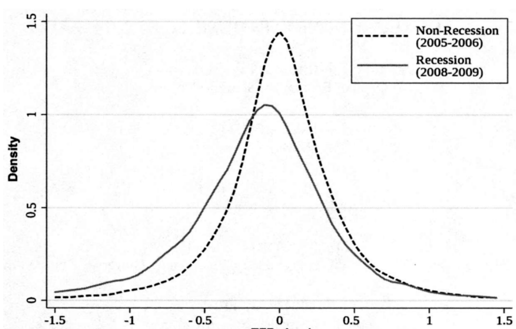

Figure 1: TFP shock density before and during recession

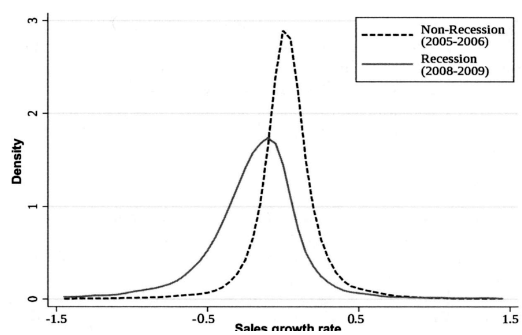

Figure 2: Sales growth density before and during recession

8.2 Figure 3 and Table I: uncertainty is higher in recessions

Read:

- Figure 3 plots the interquartile range of TFP shocks over time and shows clear countercyclicality.
- Table I reports regression evidence that standard deviation and interquartile range of TFP shocks rise significantly in recession years.
- Skewness and kurtosis are not the main drivers; the central empirical fact is a rise in dispersion.

Economic message: The baseline Census evidence supports the interpretation of recessions as negative first-moment plus positive second-moment events.

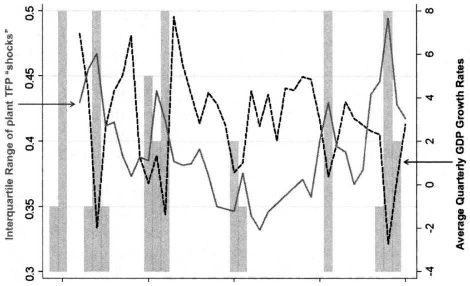

Figure 3: TFP shocks over time

TABLE I

UNCERTAINTY IS HIGHER DURING RECESSIONS[a]

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
:ndent :iable:	S.D. of log(TFP) shock	Skewness of log(TFP) shock	Kurtosis of log(TFP) shock	IQR of log(TFP) shock	IQR of output growth	IQR of sales growth	IQR of stock returns	IQR of industria prod. growth
›le:	Plants (manufact.)	Plants (manufact.)	Plants (manufact.)	Plants (manufact.)	Plants (manufact.)	Public firms (all sectors)	Public firms (all sectors)	Industries (manufact.)
ssion	0.064***	−0.248	−1.334	0.061***	0.077***	0.032***	0.025***	0.044***
	(0.009)	(0.191)	(1.994)	(0.019)	(0.019)	(0.007)	(0.003)	(0.004)
ı of dep. var.	0.503	−1.525	20.293	0.395	0.196	0.186	0.104	0.101
. GDP growth	−0.450***	0.143	0.044	−0.444***	−0.566***	−0.275***	−0.297***	−0.335***
ıency	Annual	Annual	Annual	Annual	Annual	Quarterly	Monthly	Monthly
;	1972–2011	1972–2011	1972–2011	1972–2011	1972–2011	1962:1–2010:3	1960–2010	1972–2010
rvations	39	39	39	39	39	191	609	455
:rlying sample	461,232	461,232	461,232	461,232	461,232	320,306	931,143	70,487

ach column reports a time-series ordinary least squares (OLS) regression point estimate (and standard error below in parentheses) of a measure of uncertainty on a recession indicator. Th ion indicator is the share of quarters in that year in a recession in columns 1–5, whether that quarter was in a recession in column 6, and whether the month was in recession in columns 7 and ε sions are defined using the National Bureau of Economic Research (NBER) data. In the bottom panel we report the mean of the dependent variable and its correlation with real GDP growth. I ns 1–5, the sample is the population of manufacturing establishments with 25 years or more of observations in the Annual Survey of Manufactures (ASM) or Census of Manufactures (CM) surve :n 1972 and 2009, which contains data on 15,673 establishments across 40 years of data (one more year than the 39 years of regression data since we need lagged TFP to generate a TFP shoc re). We include plants with 25+ years to reduce concerns over changing samples. In column 1, the dependent variable is the cross-sectional standard deviation (S.D.) of the establishment-leve to total factor productivity (TFP). This shock is calculated as the residual from the regression of log(TFP) at year $t+1$ on its lagged value (year t), a full set of year dummies, and establishmen :ffects. In column 2, we use the cross-sectional coefficient of skewness of the TFP shock, in column 3, the cross-sectional coefficient of kurtosis, and in column 4, the cross-sectional interquartil of this TFP shock as an outlier robust measure. In column 5, the dependent variable is the interquartile range of plants' sales growth. In column 6, the dependent variable is the interquartil of firms' sales growth by quarter for all public firms with 25 years (100 quarters) or more in Compustat between 1962 and 2010. In column 7, the dependent variable is the within firm-quarte ıartile range of firms' monthly stock returns for all public firms with 25 years (300 months) or more in Center for Research in Security Prices (CRSP) between 1960 and 2010. Finally, in colum dependent variable is the interquartile range of industrial production growth by month for manufacturing industries from the Federal Reserve Board's monthly industrial production database gressions include a time trend and for columns 1–5 Census year dummies (for Census year and for three lags). Robust standard errors are applied in all columns to control for any potentia :orrelation. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance. Results are also robust to using Newey–West corrections for the standard errors. Data ar ›le at http://www.stanford.edu/~nbloom/RUBC.zip.

Table I: Uncertainty is higher during recessions

8.3 Figure 4 and Table II: robustness and industry-level evidence

Read:

- Figure 4 shows that different ways of constructing TFP shocks deliver similar countercyclical dispersion patterns.
- Table II shows that within-industry uncertainty is higher when industry output growth is lower.
- The industry result holds with size, age, capital-labor, and geographic controls.

Economic message: The result is not driven by a single TFP specification or by aggregate composition. It appears within industries.

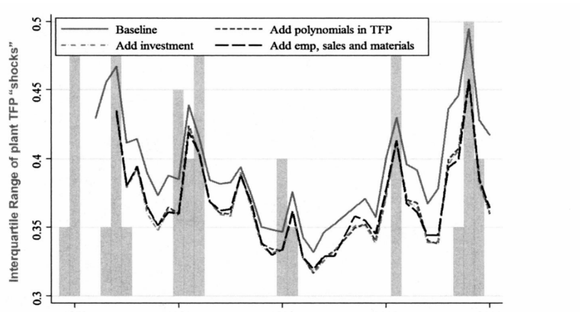

Figure 4: Robustness to different TFP shock measures

TABLE II

UNCERTAINTY IS ALSO ROBUSTLY HIGHER AT THE INDUSTRY LEVEL DURING INDUSTRY "RECESSIONS"[a]

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
:ndent variable:	IQR of establishment TFP shocks within each industry–year cell								
ification:	Baseline	Median industry output growth	IQR of industry output growth	Median industry establishment size	IQR of industry establishment size	Median industry capital/labor ratio	IQR of industry capital/labor ratio	IQR of industry TFP spread	Industry geographi spread
stry output growth	−0.132***	−0.142***	−0.176***	−0.119***	−0.116***	−0.111***	−0.111***	−0.191***	−0.133***
	(0.021)	(0.021)	(0.047)	(0.024)	(0.022)	(0.034)	(0.030)	(0.041)	(0.028)
action of industry tput growth with : variable in :cification row		0.822	0.882	−0.032	−0.033	−0.197	−0.265	0.123	0.007
		(0.630)	(0.996)	(0.038)	(0.026)	(0.292)	(0.330)	(0.084)	(0.122)
;	1972–2009	1972–2009	1972–2009	1972–2009	1972–2009	1972–2009	1972–2009	1972–2009	1972–2009
rvations	16,451	16,451	16,451	16,451	16,451	16,451	16,451	16,451	16,451
:rlying sample	446,051	446,051	446,051	446,051	446,051	446,051	446,051	446,051	446,051

ach column reports the results from an industry-by-year OLS panel regression, including a full set of industry and year fixed effects. The dependent variable in every column is the interquartil (IQR) of establishment-level TFP shocks within each SIC four-digit industry–year cell. The regression sample is the 16,451 industry–year cells of the population of manufacturing establishment 5 years or more of observations in the ASM or CM survey between 1972 and 2009 (which contains 446,051 underlying establishment years of data). These industry–year cells are weighted in th sion by the number of establishment observations within that cell, with the mean and median number of establishments per industry–year cell equal to 27.1 and 17, respectively. The TFP shoc ulated as the residual from the regression of log(TFP) at year $t+1$ on its lagged value (year t), a full set of year dummies, and establishment fixed effects. In column 1, the explanatory variabl median of the establishment-level output growth in that industry–year. In columns 2–9, a second variable is also included that is an interaction of that explanatory variable with an industry-leve :teristic. In columns 2 and 3, this is the median and IQR of industry-level output growth, in columns 4 and 5 this is the median and IQR of industry-level establishment size in employees, i ns 6 and 7, this is the median and IQR of industry-level capital/labor ratios, in column 8 this is the IQR of industry-level TFP levels (note the mean is zero by construction), while finally i n 9, this interaction is the dispersion of industry-level concentration measured using the Ellison–Glaeser dispersion index. Standard errors clustered by industry are reported in brackets belov point estimate. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table II: Industry uncertainty and recessions

8.4 Table III and Table IV: external uncertainty measures and model calibration

Read:

- Table III links plant TFP shock measures to parent firm stock-return volatility and other uncertainty proxies.
- Table IV reports calibrated model parameters, including preferences, technology, and adjustment costs.
- These tables connect the empirical uncertainty measures to financial proxies and to the quantitative model.

Economic message: Stock market volatility is a meaningful proxy for real microeconomic shocks, and the model is parameterized to match firm and macro dynamics.

TABLE III

CROSS-SECTIONAL ESTABLISHMENT UNCERTAINTY MEASURES ARE CORRELATED WITH FIRM AND INDUSTRY TIME SERIES UNCERTAINTY MEASURES[a]

	(1)	(2)	(3)	(4)	(5)
ndent variable	Mean of establishment *absolute* (TFP shock) within firm year				Mean of establishment absolute (TFP shocks within industry year
le	Establishments (in manufacturing) with a parent firm in compustat				Manufacturing industries
ession panel dimension	Firm by year				Industry by year
of parent daily stock urns within year	0.317*** (0.091)				
of parent monthly stock urns within year		0.275*** (0.083)			
of parent daily stock returns hin year, leverage adjusted			0.381*** (0.118)		
of parent quarterly sales wth within year				0.134*** (0.029)	
of monthly industrial duction within year					0.330*** (0.060)
l effects and clustering	Firm	Firm	Firm	Firm	Industry
s/industries	1838	1838	1838	1838	466
rvations	25,302	25,302	25,302	25,302	16,406
rlying observations	172,074	172,074	172,074	172,074	446,051

he dependent variable is the mean of the absolute size of the TFP shock at the firm–year level (columns 1–4) and industry–year level (column 5). This TFP shock is calculated as the residua he regression of log(TFP) at year $t+1$ on its lagged value (year t), a full set of year dummies, and establishment fixed effects, with the absolute size generated by turning all negative value e. The regression sample in columns 1–4 are the 25,302 firm–year cells of the population of manufacturing establishments with 25 years or more of observations in the ASM or CM surve en 1972 and 2009 that are owned by Compustat (publicly listed) firms. This covers 172,074 underlying establishment years of data. The regression sample in column 5 is the 16,406 industry–yea f the population of manufacturing establishments with 25 years or more of observations in the ASM or CM survey between 1972 and 2009. The explanatory variables in columns 1–3 are th l standard deviation of the parent firm's stock returns, which are calculated using the 260 daily values in columns 1 and 3 and the 12 monthly values in column 2. For comparability of monthly an alues, the coefficients and S.E. for the daily returns in columns 1 and 3 are divided by sqrt(21). The daily stock returns in column 3 are normalized by the (equity/(debt + equity)) ratio to contr erage effects. In column 4, the explanatory variable is the standard deviation of the parent firm's quarterly sales growth. Finally, in column 5, the explanatory variable is the standard deviatio industry's monthly industrial production data from the Federal Reserve Board. All columns have a full set of year fixed effects, with columns 1–4 also having firm fixed effects while column lustry fixed effects. Standard errors clustered by firm/industry are reported in brackets below every point estimate. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10 cance.

Table III: Cross-sectional uncertainty measures

TABLE IV

CALIBRATED MODEL PARAMETERS[a]

ences and Technology		
	$0.95^{1/4}$	Annual discount factor of 95%
	1	Unit elasticity of intertemporal substitution (Khan and Thomas (2008))
	2	Leisure preference, households spend 1/3 of time working
	1	Infinite Frisch elasticity of labor supply (Khan and Thomas (2008))
	0.25	CRS production, isoelastic demand with 33% markup
	0.5	CRS labor share of 2/3, capital share of 1/3
	0.95	Quarterly persistence of aggregate productivity (Khan and Thomas (2008))
	0.95	Quarterly persistence of idiosyncratic productivity (Khan and Thomas (2008)
tment Costs		
	2.6%	Annual depreciation of capital stock of 10%
	8.8%	Annual labor destruction rate of 35% (Shimer (2005))
	0	Fixed cost of changing capital stock (Bloom (2009))
	33.9%	Resale loss of capital in % (Bloom (2009))
	2.1%	Fixed cost of changing hours in % of annual sales (Bloom (2009))
	1.8%	Per worker hiring/firing cost in % of annual wage bill (Bloom (2009))

he model parameters relating to preferences, technology, and adjustment costs are calibrated as referenced above.

Table IV: Calibrated model parameters

8.5 Table V and Table VI: estimated uncertainty process

Read:

- Table V reports estimated uncertainty parameters. High uncertainty is persistent, and both macro and micro volatility rise in high-uncertainty states.
- Table VI compares data and model moments for macro and micro uncertainty.
- The model reproduces the mean, dispersion, skewness, and serial correlation of uncertainty reasonably well.

Economic message: The model's uncertainty process is not chosen arbitrarily; it is estimated to match empirical moments of uncertainty.

TABLE V

ESTIMATED UNCERTAINTY PARAMETERS[a]

Quantity	Estimate	Standard Error	
σ^A_L	0.67	(0.098)	Quarterly standard deviation of macroproductivity shocks (%)
σ^A_H/σ^A_L	1.6	(0.015)	Macrovolatility increase in high uncertainty state
σ^Z_L	5.1	(0.807)	Quarterly standard deviation of microproductivity shocks (%)
σ^Z_H/σ^Z_L	4.1	(0.043)	Microvolatility increase in high uncertainty state
$\pi^\sigma_{L,H}$	2.6	(0.485)	Quarterly transition probability from low to high uncertainty (%)
$\pi^\sigma_{H,H}$	94.3	(16.38)	Quarterly probability of remaining in high uncertainty (%)

[a]The uncertainty process parameters are structurally estimated through an SMM procedure (see the main text and Appendix C in the Supplemental Material). The estimation process targets the time-series moments of the cross-sectional interquartile range of the establishment-level shock to estimated productivity in the Census of Manufactures and Annual Survey of Manufactures manufacturing sample, along with the time-series moments of estimated heteroskedasticity of the U.S. aggregate Solow residual based on a GARCH(1, 1) model. Both sets of target moments from the data are computed from 1972 to 2010.

Table V: Estimated uncertainty parameters

when an uncertainty shock arrives. Idiosyncratic volatility is estimated to equal 5.1% and

TABLE VI

UNCERTAINTY PROCESS MOMENTS[a]

	Data	Model
Macro-moments		
Mean	3.36	3.58
Standard deviation	0.76	0.59
Skewness	0.83	1.18
Serial correlation	0.88	0.83
Micro-moments		
Mean	39.28	38.44
Standard deviation	4.89	4.55
Skewness	1.16	0.81
Serial correlation	0.75	0.65

[a]The microdata moments are calculated from the U.S. Census of Manufactures and Annual Survey of Manufactures sample using annual data from 1972–2010. Microdata moments are computed from the cross-sectional interquartile range of the estimated shock to establishment-level productivity, in percentages. The model micro-moments are computed in the same fashion as the data moments, after correcting for measurement error in the data establishment-level regressions and aggregating to annual frequency. The macrodata moments refer to the estimated heteroskedasticity from 1972–2010 implied by a GARCH(1, 1) model of the annualized

Table VI: Uncertainty process moments

8.6 Table VII and Figure 5: business-cycle moments and firm thresholds

Read:

- Table VII shows that the model generates business-cycle statistics that are close to the data for output, investment, consumption, and hours.
- Figure 5 shows how uncertainty widens the inaction region for hiring and firing.
- Higher uncertainty moves thresholds outward and leaves more firms inactive.

Economic message: The micro real-options mechanism–firms waiting near action thresholds–is the foundation for the aggregate output decline.

TABLE VII

BUSINESS CYCLE STATISTICS[a]

	Data			Model		
	$\sigma(x)$			$\sigma(x)$		
	$\sigma(x)$	$\sigma(y)$	$\rho(x,y)$	$\sigma(x)$	$\sigma(y)$	$\rho(x,y)$
Output	1.6	1.0	1.0	2.0	1.0	1.0
Investment	7.0	4.5	0.9	11.9	6.0	0.9
Consumption	1.3	0.8	0.9	0.9	0.4	0.5
Hours	2.0	1.3	0.9	2.4	1.2	0.8

[a]The first panel contains business cycle statistics for quarterly U.S. data covering 1972Q1–2010Q4. The term $\sigma(x)$ is the standard deviation of the log variable, $\sigma(y)$ is the standard deviation in log variable relative to the standard deviation of log output, and $\rho(x,y)$ is the correlation of the log variable and log output. All business cycle data are current as of July 14, 2014. Output is real gross domestic product (FRED GDPC1), investment is real gross private domestic investment (GPDIC1), consumption is real personal consumption

Table VII: Business-cycle statistics

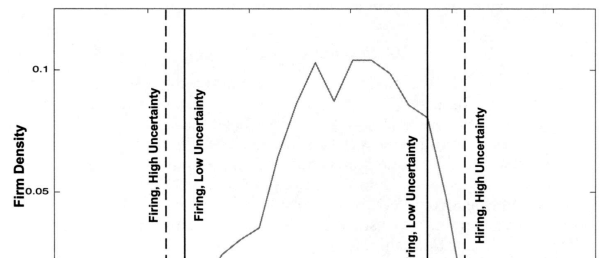

Figure 5: Hiring and firing thresholds

8.7 Figure 6 and Figure 7: output drop, labor drop, investment drop, and misallocation

Read:

- Figure 6 shows that a pure uncertainty shock lowers output sharply on impact by about 2.5 percent.
- Figure 7 decomposes the response into labor, investment, consumption, and labor misallocation.
- Labor and investment fall immediately, while misallocation rises as productive and unproductive firms delay adjustment.

Economic message: Uncertainty shocks affect aggregate output through inaction and misallocation, not merely through a decline in average productivity.

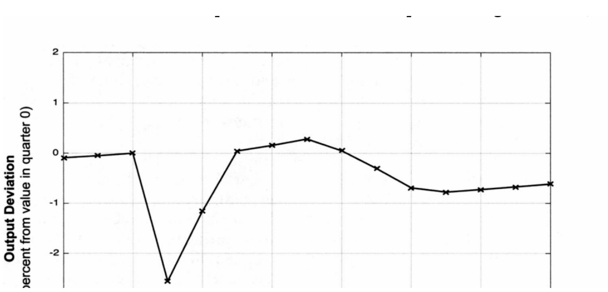

Figure 6: Output response to an uncertainty shock

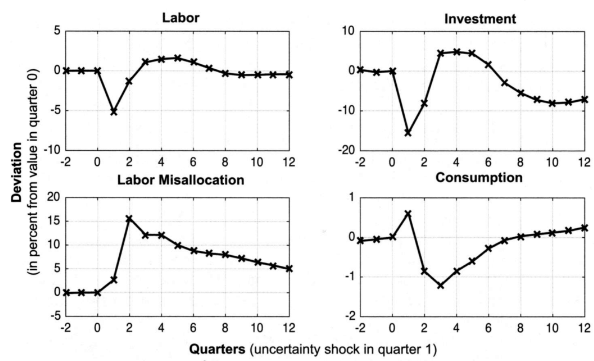

Figure 7: Labor, investment, consumption, and misallocation

8.8 Figure 8 and Figure 9: adding first moments and robustness

Read:

- Figure 8 adds a negative first-moment shock to the uncertainty shock and shows a deeper, more realistic recession.
- Figure 9 shows robustness to alternative calibrations of uncertainty parameters and adjustment costs.
- The output, labor, investment, and consumption patterns are broadly preserved.

Economic message: Second-moment shocks matter, but recession-like dynamics are best explained by a combination of negative first-moment and positive second-moment shocks.

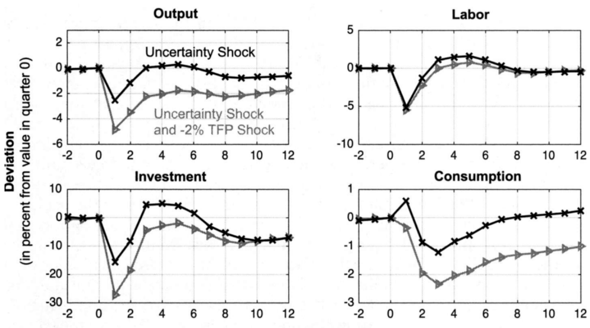

Figure 8: Adding a negative first-moment shock

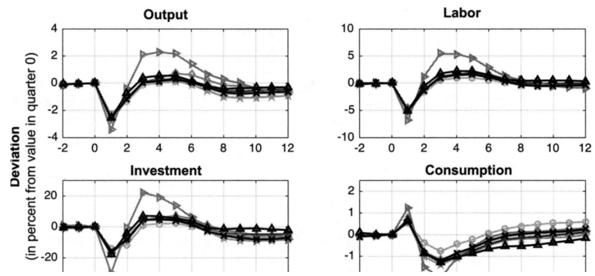

Figure 9: Robustness of uncertainty shock effects

8.9 Figure 10 and Figure 11: decomposition and laws of motion

Read:

- Figure 10 decomposes the effects into general equilibrium, partial equilibrium with adjustment costs, and partial equilibrium without adjustment costs.
- Without adjustment costs, the Oi-Hartman-Abel effect can raise output after uncertainty shocks.
- Figure 11 shows that lower depreciation and attrition rates attenuate but do not overturn the baseline mechanism.

Economic message: Adjustment costs are the key reason uncertainty shocks contract output rather than expand it.

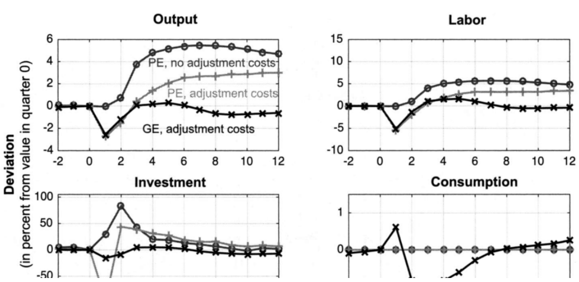

Figure 10: Decomposing GE, PE, and adjustment-cost effects

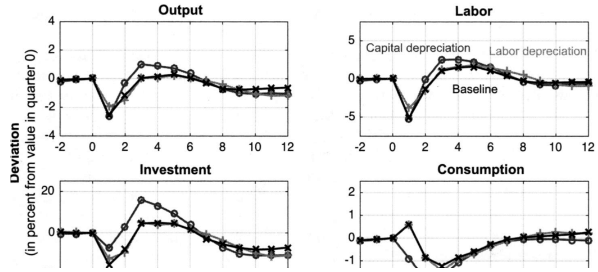

Figure 11: Depreciation and attrition robustness

8.10 Figure 12: policy is less effective when uncertainty is high

Read:

- Figure 12 compares a wage subsidy in normal times with the same subsidy during an uncertainty shock.
- The output impact of the subsidy falls by more than two thirds when uncertainty is high.
- The policy works less because fewer firms are near the margin of adjusting employment.

Economic message: Uncertainty changes not only the level of activity but also the multiplier of first-moment policies.

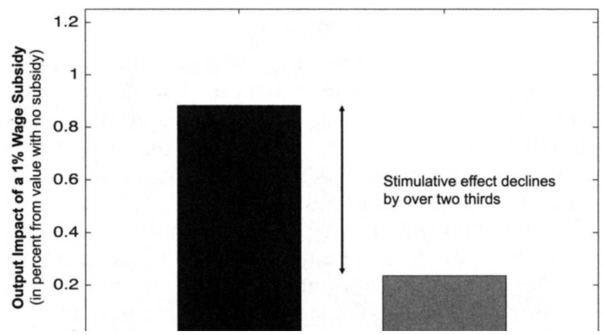

Figure 12: Wage subsidy effectiveness in normal and high-uncertainty states

9 Short summary

- Microeconomic uncertainty rises sharply in recessions.
- Plant-level TFP shock dispersion is strongly countercyclical in Census data.
- Sales, output, industry-level, and stock-return measures confirm the same pattern.
- A DSGE model with heterogeneous firms and adjustment costs can translate uncertainty shocks into large aggregate contractions.
- The key mechanism is real-options inaction: firms delay hiring, firing, investment, and disinvestment.
- A pure uncertainty shock lowers output but does not fully match consumption dynamics.
- Recession-like simulations require both a negative first-moment shock and a positive second-moment shock.
- Uncertainty reduces the effectiveness of wage subsidies because firms become less responsive to price incentives.
- The paper's key message is that business cycles involve both first-moment and second-moment shocks.

10 Conclusion

10.1 Core conclusion

Bloom, Floetotto, Jaimovich, Saporta-Eksten, and Terry show that uncertainty rises in recessions and that such increases can have quantitatively important effects in a general equilibrium model with heterogeneous firms and adjustment frictions. The paper demonstrates that uncertainty shocks can create large temporary declines in output, labor, and investment.

10.2 Economic intuition

Higher uncertainty raises the option value of waiting. Because hiring, firing, investment, and disinvestment are costly, firms delay action when volatility rises. This delay generates a fall in aggregate activity and a rise in misallocation. However, uncertainty alone does not explain all recession dynamics, especially consumption, so realistic recessions require both higher uncertainty and bad first-moment shocks.

10.3 Takeaway

The central lesson is that recessions are not only about lower expected productivity. They are also about a wider distribution of microeconomic outcomes. Policy analysis must therefore account for the fact that high uncertainty makes firms less responsive to ordinary stimulus.